

## SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

*Sub-Strand 1.1: Real Numbers — Teacher Reference (Pre-filled)*

| SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10 |                                                                                                                                                                                   |
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| Sub-Strand                                    | Sub-Strand 1.1: Real Numbers                                                                                                                                                      |
| Driving Question                              | DRIVING QUESTION: Why do some numbers from our everyday Kenyan lives "end neatly" while others go on forever — and how does knowing the difference help us make better decisions? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                        | What did I observe?                                                                                                                                                                                                                                                                                                | What did I learn?                                                                                                                                                                                                                                                                                                                                         | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Lesson 1</b><br><b>Anchoring Phenomenon &amp; Number Classification — What Kind of Number Is That?</b> | Sister Wanjiku's calculator showed $(36.8 + 37.0 + 37.5) \div 3 = 37.1$ (a decimal that terminates), but $\sqrt{(1^2+1^2)} = \sqrt{2} = 1.41421356\dots$ (a decimal that never ends and never repeats). All six health-centre measurements could be written as fractions with integer numerators and denominators. | A rational number is any number that can be written as $p/q$ where $p$ and $q$ are integers and $q \neq 0$ . All of Sister Wanjiku's health-centre measurements are rational. $\sqrt{2}$ — the tile diagonal — appears to be irrational because no fraction $p/q$ exactly equals it, though we have not yet proved why.                                   | The Real Number system branches into Rational numbers (which include integers, fractions, and terminating or repeating decimals) and Irrational numbers (which cannot be expressed as $p/q$ and whose decimals neither terminate nor repeat). Sister Wanjiku's average temperature is rational; the waiting-room tile diagonal is irrational — and understanding WHY is the driving question for this sub-strand. |
| <b>Lesson 2</b><br><b>Observe: Rational vs. Irrational Numbers — Sorting Kenyan Real-Life Data</b>        | Learners examined a curated set of ten Kenyan real-life numbers (maize prices, athletics times, tile diagonals, blood pressure values, body temperatures, $\pi$ ) and attempted to express each as a fraction $p/q$ .                                                                                              | Numbers that can be written as $p/q$ (where $p$ and $q$ are integers and $q \neq 0$ ) are rational; numbers such as $\sqrt{2}$ and $\pi$ — which produce non-terminating, non-repeating decimals — cannot be written this way and are irrational. The health volunteer's temperature average is rational; the tile diagonal ( $\sqrt{2}$ ) is irrational. | The class constructed a Claim-Evidence-Reasoning board sort and revised their real number system model to show two branches (rational and irrational) with Kenyan examples in each, directly connecting back to why the health volunteer's calculator 'ended neatly' for one calculation but 'went on forever' for another.                                                                                       |
| <b>Lesson 3</b><br><b>Explain: Defining and Formalising Rational vs. Irrational Numbers</b>               | In Lessons 1 and 2, we observed that the health volunteer's temperature average ( $37.1^\circ\text{C}$ ) produced a terminating decimal on the calculator, while the tile diagonal ( $\sqrt{2}$ ) produced a never-ending, non-repeating decimal. We also observed that $22/7$                                     | In Lesson 3, we learned the formal definitions: a RATIONAL number can always be written as $p/q$ where $p$ and $q$ are integers and $q \neq 0$ — its decimal either terminates or repeats. An IRRATIONAL number cannot be written in this form — its                                                                                                      | Using these definitions, we can now explain the health volunteer's puzzle: her temperature readings ( $36.8 = 184/5$ , $37.0 = 37/1$ , $37.5 = 75/2$ ) are all rational — they are exact ratios of integers, so they terminate on the calculator. The tile                                                                                                                                                        |

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|                                                                                                                              | produces a repeating decimal (3.142857) while $\pi$ produces a non-repeating one.                                                                                                                                                                                                                                                                                                                                                                                                     | decimal is non-terminating and non-repeating. We learned that $22/7$ is RATIONAL (it equals a fraction of integers) while $\pi$ is IRRATIONAL, even though they are close in value.                                                                                                                                                                                                                                                                                                                                                                                                                  | diagonal ( $\sqrt{2}$ ) is irrational — no ratio of integers equals it exactly — so the calculator display never ends. The difference is not about the calculator; it is about the nature of the number itself.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Lesson 4</b><br><b>Combined Operations on Rational Numbers I — Addition &amp; Subtraction in the Market and Classroom</b> | We observed that when Mama Akinyi's market prices (Ksh 45.50 + Ksh 12.75 + Ksh 38.00) and the Kisumu school stationery quantities were added and subtracted, every result was a number that could still be written as a fraction $p/q$ — the answers always 'ended neatly' or repeated, never producing a non-terminating, non-repeating decimal like $\sqrt{2}$ .                                                                                                                    | We learned that rational numbers are closed under addition and subtraction: if $a = p/q$ and $b = m/n$ are both rational, then $a + b = (pn + mq)/(qn)$ is also rational. BODMAS governs the order of combined operations. The closure property means that no matter how many rational numbers Sister Wanjiku adds or subtracts, her result will always remain rational.                                                                                                                                                                                                                             | This helps explain the anchoring phenomenon: Sister Wanjiku's average temperature ( $36.8 + 37.0 + 37.5 \div 3$ ) involves only addition and division of rational numbers. Because rational numbers are closed under addition (and, as Lesson 5 shows, under division), the result must be rational — it terminates or repeats. The tile diagonal ( $\sqrt{2}$ ), by contrast, is irrational and was never produced by adding or subtracting rational numbers; it came from a square root operation on a non-perfect square, which takes us outside the rational number set entirely. |
| <b>Lesson 5</b><br><b>Combined Operations on Rational Numbers II — Multiplication &amp; Division</b>                         | Learners observed four fully worked multiplication and division problems in a gallery walk — a Kericho tea farmer dividing $3\frac{3}{4}$ kg among 5 co-operative members, a Rift Valley maize farmer multiplying $2\frac{2}{5}$ kg/acre over $3\frac{1}{2}$ acres, a Kisumu fish trader finding $\frac{2}{3}$ of $4\frac{1}{2}$ crates, and a Mombasa coconut seller sharing $6\frac{3}{5}$ kg among 4 buyers — and annotated each solution for correctness and alternative methods. | Learners learned that multiplying or dividing rational numbers (including mixed numbers) always produces a rational number; that the correct procedure requires converting mixed numbers to improper fractions first, and inverting the divisor before multiplying in division problems; and that every agricultural result in today's lesson 'ended neatly', just like the health volunteer's average temperature.                                                                                                                                                                                  | Learners explained, through gallery-walk annotations and cold-call defences using Claim–Evidence–Reasoning, why each step in a worked solution was correct or incorrect; they consolidated the standard method for multiplying and dividing mixed numbers; and they updated their running Real Numbers Model to record that all four arithmetic operations on rational numbers yield rational results, deepening the distinction between rational and irrational numbers anchored in the driving question.                                                                            |
| <b>Lesson 6</b><br><b>Reciprocals of Real Numbers — What Happens When You "Undo" a Number on a Calculator?</b>               | Learners used calculators to find the reciprocal of every number in the health-centre data set — body temperatures (36.8, 37.0, 37.5), blood pressure values (120, 80, 135, 90), and the irrational tile diagonal $\sqrt{2}$ — and recorded that in every case the product $x \times (1/x)$ equalled exactly 1.                                                                                                                                                                       | The reciprocal (multiplicative inverse) of any non-zero real number $x$ is $1/x$ , and $x \times (1/x) = 1$ always holds. The reciprocal of a rational number is rational (e.g., $1/37.1 = 10/371$ ). The reciprocal of an irrational number is also irrational — proved by contradiction: if $1/\sqrt{2}$ were rational, $\sqrt{2}$ would be rational, contradicting Lesson 3. Zero has no reciprocal. A calculator display that 'ends' does not prove a decimal terminates — responsible calculator use requires knowing the difference between a truncated display and a truly terminating value. | Sister Wanjiku's health-centre numbers — temperatures and blood pressure readings — are all rational, so their reciprocals are also rational and 'end neatly'. The tile diagonal $\sqrt{2}$ is irrational, and its reciprocal $1/\sqrt{2}$ is also irrational — it never ends and never repeats, even though a calculator makes it look finite. This means the rational/irrational boundary is preserved under the reciprocal operation, just as Lessons 4 and 5 showed it is preserved under addition, subtraction, multiplication, and division.                                    |

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| <b>Lesson 7</b><br><b>Real Numbers in Real-Life Applications &amp; DQB Consolidation</b>                                | Learners computed the mean body temperature of three Mathare North Health Centre patients (Tasks A), calculated total and unit githeri ingredient costs for a 200-learner school canteen in Kisumu (Task B), and determined that $\sqrt{5}$ m (a Rift Valley farm fence-post length) is irrational, approximating it to 4 significant figures (Task C).                                                                                                                                                                                                                                                                                               | Learners confirmed that numbers arising from everyday Kenyan contexts — health temperatures, food costs, masses — are rational because they can be expressed as exact fractions, giving terminating or repeating decimals; whereas numbers arising from square roots of non-perfect squares (like $\sqrt{2}$ and $\sqrt{5}$ ) are irrational, producing non-terminating non-repeating decimals. The real-number system contains both types.                                                                                                                                                                                                                                                                                                                                                                                                                               | Learners explained, through group presentations and targeted cold-calling, why the Mathare North nurse's average temperature 'ended neatly' (rational arithmetic on terminating decimals always yields a rational result) while the tile-diagonal calculation displayed a never-ending decimal ( $\sqrt{2}$ is irrational). They updated their DQB and personal Real-Number Models to consolidate evidence from all seven lessons.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Lesson 8</b><br><b>Model Building, Synthesis &amp; Final Explanation — Completing the Real Numbers Reference Map</b> | Learners observed their own Lesson-1 naive explanations alongside their Lesson-8 precise mathematical explanations, making their conceptual growth visible. They observed that the Reference Map — populated entirely with Kenyan-context numbers from health, agriculture, and everyday life — forms a coherent nested hierarchy ( $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ , with irrationals alongside $\mathbb{Q}$ inside $\mathbb{R}$ ). They also observed, through the DQB closure, that every question the class asked on Day 1 about Sister Wanjiku's calculator display now has a precise mathematical answer. | Learners learned that the real-number system is a complete, nested structure where every number encountered in everyday Kenyan life — body temperatures, shoe sizes, blood pressure ratios, market prices, tile diagonals — can be precisely located. Rational numbers (expressible as $p/q$ , $q \neq 0$ ) produce terminating or repeating decimals and are closed under all four arithmetic operations. Irrational numbers ( $\sqrt{2}$ , $\pi$ , $\sqrt{5}$ ) cannot be expressed as $p/q$ and produce non-terminating, non-repeating decimals. The reciprocal of any non-zero real number $x$ is $1/x$ with $x \times (1/x) = 1$ . Knowing which type of number you are working with determines whether a calculation 'ends neatly' or requires an approximation — a distinction with direct consequences for decision-making in health, construction, and commerce. | The phenomenon is now fully explained: Sister Wanjiku's temperature average $(36.8 + 37.0 + 37.5) \div 3 = 111.3 \div 3 = 37.1$ ends neatly because all three temperatures are rational numbers (e.g., $36.8 = 368/10$ ), and rational numbers are closed under addition and division by a non-zero integer — the result $37.1 = 371/10$ is itself rational, with a terminating decimal. The tile diagonal $\sqrt{2}$ goes on forever because 2 is not a perfect square: no fraction $p/q$ (with $p, q \in \mathbb{Z}$ , $q \neq 0$ ) satisfies $(p/q)^2 = 2$ exactly, so $\sqrt{2}$ is irrational and its decimal is non-terminating and non-repeating. Knowing this difference helps Sister Wanjiku — and any Kenyan decision-maker — understand when a calculator result is exact and when it is merely an approximation that must be treated with appropriate caution. |

## ATTRIBUTION AND LICENSING

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